HDOC- 3-1 – 1A

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Program to Swap Two Numbers Using a Third Variable

This document provides a clean, well-commented implementation for swapping two numbers using a third variable in C. The program takes two numbers as input, uses a temporary variable to exchange their values, and displays the swapped numbers.

1. C Implementation

Step 1: Understand the Logic

• A third variable (temp) is used to temporarily store the value of the first number.

• The value of the second number is then assigned to the first number.

• Finally, the value stored in the third variable is assigned to the second number.

Step 2: Define Variables and Data Types

• Use integer variables (int) to store the two numbers and the temporary variable.

Step 3: Take User Input

• Read two numbers using scanf("%d %d", &num1;, &num2;).

Step 4: Swap the Values

1. Store num1 in temp.

2. Assign num2 to num1.

3. Assign temp to num2.

Step 5: Format the Output

• Print the values of the two numbers before and after swapping.

Complete C Code

```
/*
 * Q3 (User Inputs, Operations & Output)
 * Write a program to swap two numbers using a third variable.
 */

#include <stdio.h>

int main() {
    int num1, num2, temp;

    // Step 1: Read two numbers
    scanf("%d %d", &num1, &num2);

    // Step 2: Display values before swapping
    printf("Before swapping: num1=%d, num2=%d\n", num1, num2);

    // Step 3: Swap using a third variable
    temp = num1;
    num1 = num2;
```

```
    num2 = temp;

    // Step 4: Display values after swapping
    printf("After swapping: num1=%d, num2=%d\n", num1, num2);

    return 0;
}
```

Dry Run with Test Cases

Sample Test Case 1:

• Input: 10 20

• Calculation:

o temp = 10

o num1 = 20

o num2 = 10

• Output: Before swapping: num1=10, num2=20

After swapping: num1=20, num2=10

Sample Test Case 2:

• Input: 5 15

• Calculation:

o temp = 5

o num1 = 15

o num2 = 5

• Output: Before swapping: num1=5, num2=15

After swapping: num1=15, num2=5